Spotted Lanternfly Eradication via Mechanical Rotating Trap

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Context and Problem Statement:

The New York Grape industry is suffering from a new invasive pest, the Spotted Lanternfly (SLF). SLF are swarming vineyards, weakening and killing grapevines. The *Spotted Lanternfly Mechanical Grape Harvester: Initial Study* counted an average of 22.9 SLF per vine (about 510 SLF per 100-foot row). Furthermore, any SLF that gets into the harvest bins is considered foreign matter, raising issues with the FDA. The industry is looking for ways to remove SLF from the vines and eliminate their matter from the harvest. Two obvious solutions to preventing SLF contamination are spraying insecticide shortly before harvest and removing/killing the SLF during harvesting. These present a few issues: Using mass pesticides might cause environmental problems, and pesticide use involves a complicated set of Food Safety Requirements from the USDA. Killing while harvesting potentially would incur damage to the grape crops while being harvested. Our team focused on the sub-problem of directly killing SLF in a 1m radius of the grape vines using a passive, low-maintenance mechanical device. Our target success criterion is eliminating at least 90% of SLF within a 1m radius, with no reliance on pesticides and minimal labor for the farmers.

Final Prototype and Application:

We designed and built a solar-powered, rotating mechanical SLF trap that is intended to be placed near the grapevines throughout the vineyard. The device is a compact cylindrical unit which is approximately 10 inches in diameter and 9 inches tall with three sections: an electronics bay at the base, a bottom drum that serves as the kill chamber, and a top drum that houses the rotating rotor assembly. SLF are lured into the open upper chamber using Tree of Heaven sap as a natural attractant, placed on the bait dishes underneath the rotor. A solar-powered 6V DC motor (100 RPM, speed-controlled down to 5–10 RPM) drives a three-blade rotor that sweeps the SLF into a one-way opening into the sealed bottom drum, which is filled with soapy water mixed with essential oils to kill them. The upper chamber, lower chamber, and electronics bay each separate cleanly from one another for easy maintenance; the growers simply have to detach, rinse, refill, and place. This solution is scalable; many traps can be placed throughout the vineyard for broad coverage.

Conclusion and Recommendation:

The mechanical trap is ready for field testing - it is functional and demonstrably easy to use and implement. As a prototype, the design has achieved the success criteria of being small enough to handle, simple enough to deconstruct and reconstruct in the field, and cheap enough to produce en masse. To reiterate, the main purpose of our device is to reduce the amount of spotted lantern flies that infest grape vineyards around a 1-meter radius, which would increase the rate of grape production for the user. Our most recent prototype proves that the mechanical and electrical core of the rotating SLF trap works. All three physical constraints were met with a significant margin: the prototype weighs 0.976 kg (target < 9 kg), occupies 0.014406 m^3 (target < 0.028m^3), and can be assembled in under 2 minutes. The motor-driven rotation at 5 rpm was confirmed to be continuous and stable under load, validating the core sweep-and-trap mechanism. Solar voltage output has been confirmed at 6V, though there is a downstream wiring fault preventing full solar-to-motor connectivity. Our current design also relies on the board being

powered via USB-C, which works, but in the future, the device should work wirelessly through an internal power supply(Li-Po Battery).
The main unknowns lie in the areas of testing regarding actual flies, durability under outdoor conditions, and effectiveness versus other methods. Due to its cheap nature and minimal profile, while it may not be a game-changer in SLF eradication, it is at least a promising background implementation that can reduce SLF numbers in the field at low cost to the vineyard. This conclusion is derived from the following test results, which are further explained below: the 6V DC motor is capable of producing a continuous sufficient-torque output for an extended period of time, and the solar capacity is demonstrably possible but requires further testing in outdoor lighting conditions with the proper LiPo battery setup.
That being said, this design has much room for improvement and many areas that could be explored further. For example, one area that could be tested is the effectiveness of the attractant used to bait the SLF. The attractant that was decided on initially was Tree of Heaven sap, since it was established that it would be effective at baiting SLF, but there could be other, better, cheaper alternatives. Another area that could be explored is the number of fins and chambers on the spinner assembly. Currently there are 6 chambers and 3 spinners, but it is unsure if it is the most optimal setup. Since the prototype is quite simple to manufacture and assemble, an effective testing plan could be to have multiple of these devices out on a field with different setups in order to test multiple variables. The current design could act as a control group, and subsequent designs implementing the changes mentioned could be deployed, in which the results could be compared to the control. A proper measure of success would be the number of SLF collected in a certain period of time, where the setup with the most lantern flies would be considered the more efficient one. One problem that could arise with this testing plan would be external variables depending on the physical locations of these prototypes. If they are placed too far apart from each other, it would be difficult to determine the effectiveness, since some locations may have more SLF than others. In order to counteract this, the prototypes would need to be placed in close proximity to each other.

Testing and Results:

Test 1: Motor & Power source

This test is focused on the motor and an arbitrary power source to check if the battery is able to drive the motor and the motor is able to drive the rotation of the system (the rotor). This test is accomplished by assembling our system (connecting the motor to a power source and switch, connecting it to an axle and secondarily to the rotor, all constrained by the container components of the trap) and flipping the switch, then checking to see if the system rotates as expected. The test results were conclusive and positive - we were able to drive the motor at ~100 rpm using a 6V power supply, which turned the motor shaft alone. When coupled with the extended shaft and the rotor, the motor supplied enough torque to drive the rotor at the desired speed (100 rpm). Wear and tear have been accounted for by minimizing friction and operating at low speeds (using a ball bearing and a slow rpm DC motor). Furthermore, we confirmed that the motor can operate continuously - the expected function in the field. For the next iteration, we will optimize the motor mount, axle connections, and rotor path (eliminate unnecessary friction) to ensure smooth operation. We will also test different RPMs using the speed controller.

Test 2: Solar panels

This test is purely focused on the solar panels (the cells), checking that they are able to generate enough voltage to power our system. There are 2 stages of this set of testing: solar panel voltage output (initial test), and capacity for the panel to drive the motor (secondary test). The primary/initial test is accomplished by hooking up a multimeter to the solar cell-power converter system to ensure at least a 6V

output, while the cell is lit by sunlight. The secondary test is accomplished by disconnecting the system from the battery source and taking it outside into the sun/well lit window area, and checking to see if the motor rotates. The primary test results were conclusive and positive. When the multimeter was connected to the solar cell-power converter assembly, it gave a reading of 6V, which is enough to power our 6V DC motor. The secondary test was inconclusive - we were unable to directly power the motor using solar power. However, this test showed that the issue with the solar power component was downstream from the cell itself and the power converter. For the next iteration, it may be worthwhile to consider a simple battery-powered option if the solar power continues to be finicky. We will continue to experiment with troubleshooting the connectivity downstream from the solar cell, but will consider pivoting to a chargeable/replaceable battery pack to power the motor instead (using a 6V battery pack).

Prototype and Testing Details:

Our design went through multiple iterations of prototyping, each working towards our overall design success criteria, but also permuted with the intent of solving certain test cases.

Our first prototype was a cardboard mock-up, built for us to answer questions about motor placement, ease of assembly, and size and weight operability. The goal was not to build a functional trap, but to model the physical components to qualitatively check for operation feasibility and for obvious physical constraints. We learned that the best place to store the motor and power source is in the basin underneath the trap. The design does excel in ease of assembly via “stackability”, and the design is neither too heavy nor too bulky to carry.

Our second, “functional” prototype allowed us to perform the central tests listed and detailed above. We ordered electronics from McMaster and Amazon, and 3D printed the rest of the infrastructure. The result was a lightweight but sturdy model that most closely resembled our final design intent, and allowed us to perform effective testing on the electronics while remaining within the physical size regime established by the cardboard mock-up. Our model features a bottom-up assembly, with the following “stackable” order: Electronics bay - Bottom Drum - Top Drum - Rotor. Internally, all electronics except for the motor are stored in compartments within the electronics bay. The bay, bottom drum, and top drum “stack” onto each other using gravity to constrain z translation, and an internal constraint extrusion to limit x and y translation. The motor screws into the bottom drum’s motor housing compartment, with the axle sticking upwards. This is connected to the main axle (press fit), which will connect to the rotor (hex joint, with a press fit). The upper connection passes through the hole in the center of the top drum, such that the rotor is vertically supported by the inner ring of the ball bearing fixed to the drum. The motor drives the axle, which simply drives the rotor. All of the 3D printed parts were modeled in CAD using parametric modeling, and they are all concentric circles or patterns. The careful preparation of a sturdy foundation, which effectively functioned as expected, save for some tolerance issues, allowed the motor and electronics testing (described above) to be performed quantitatively.

Cost Analysis:

In terms of cost effectiveness of the prototype, the total cost of the components necessary to assemble it is around $92.05. While this is under the budget of $350 by more than a third, there are still parts that could be cheaper. Since the resources were limited to McMaster and Amazon, some of the components were more expensive than other suppliers. Most of the electronics could be found on Aliexpress for a substantially lower cost.

Bill of Materials (Parts List):

Part Name	McMaster Code/Amazon Link	Specs	Fabrication	Cost (Per Order)
Bearing	60355K178	Ball, ⅝” shaft dia, 1 ⅜” housing dia, steel	N/A	$8.20
Switch	7343K184-7343K186	28V DC, Maintained, 2 Terminals (On/Off), Screw Terminals	N/A	$9.66
Solar Panel (3x)	Solar Panels	4.3”x2.4” cells, 11.8” cable, DC 6V 1W.	N/A	$15.59
Solar Power Adapter	Power Adapter	Type C, Solar - MCU	N/A	$18.23
Boost Converter	6V Boost Converter	2V-24V to 5V-28V, pack of 10 (1 needed)	N/A	$6.99
Speed Controller	Speed Controller	1.8-12V, DC	N/A	$7.99
Motor	Motor	DC 6V 5rpm Gear Motor, pack of 3 (1 needed)	N/A	$9.99
Rotor	N/A - Self-Fabricated	240mm dia, 30mm height, 3 blades, patterned extrusions on underside	3D Printed	$2.43 (Estimated Filament Cost)
Top Drum	N/A - Self-Fabricated	254 mm dia, 60mm height. Centered hole 60mm dia,	3D Printed	$7.82 (Estimated Filament Cost)
Bottom Drum	N/A - Self-Fabricated	254mm dia, 125 mm height	3D Printed	$9.93 (Estimated Filament Cost)
Electronics Bay	N/A - Self-Fabricated	254mm dia, 45 mm height	3D Printed	$5.56 (Estimated Filament Cost)
Shaft	N/A - Self-Fabricated	Couples to 3mm motor and 15mm rotor hex, with 10 mm main shaft (195mm total length)	3D Printed	$0.18 (Estimated Filament Cost)

Cost for a Single Assembly (calculated using individual component cost): $92.05